

IGNOU BCS-011: Computer Basics and PC Software

Term-End Examination: June 2020

Complete Solved Question Paper in LaTeX

Contents

Paper Pattern	2
1 Question 1	2
2 Question 2	6
3 Question 3	8
4 Question 4	11
5 Question 5	12
Final Exam Note	14

Paper Pattern

- Time: 3 Hours
- Question No. 1 is compulsory and carries 40 marks.
- Attempt any three questions from the rest. For preparation, this solution answers every question in the paper.

Maximum Marks: 100

1 Question 1

1(a) Convert the following as asked.

Answer

(i) Convert $(0.375)_{10}$ to binary.

Multiply the fractional part repeatedly by 2:

$$0.375 \times 2 = 0.75 \Rightarrow 0, \quad 0.75 \times 2 = 1.5 \Rightarrow 1, \quad 0.5 \times 2 = 1.0 \Rightarrow 1.$$

Therefore,

$$(0.375)_{10} = (0.011)_2.$$

(ii) Convert $(1010\ 1101)_2$ to decimal.

$$\begin{aligned}(10101101)_2 &= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 128 + 32 + 8 + 4 + 1 = 173.\end{aligned}$$

Thus,

$$(1010\ 1101)_2 = (173)_{10}.$$

(iii) Convert $(F2B)_{16}$ to binary.

Convert each hexadecimal digit into 4 binary bits:

$$F = 1111, \quad 2 = 0010, \quad B = 1011.$$

Hence,

$$(F2B)_{16} = (1111\ 0010\ 1011)_2.$$

(iv) Convert $(1001\ 1101)_2$ to hexadecimal. The scanned paper shows a typographical/OCR ambiguity as $(1001\ 1401)_2$; since 4 is not a binary digit, the valid binary form is taken as $(1001\ 1101)_2$.

Group bits in fours from the right:

$$1001\ 1101 \Rightarrow 1001 = 9, \quad 1101 = D.$$

Therefore,

$$(1001\ 1101)_2 = (9D)_{16}.$$

1(b) Why does a computer need RAM although CPU processes data and I/O devices perform input/output?

Answer

RAM is needed because the CPU cannot efficiently work directly with slow secondary storage or only with input/output devices. RAM acts as the working area of the computer.

- It stores the operating system, running programs and currently used data.
- It provides very fast access compared with hard disk, SSD, pen drive or optical disk.
- It temporarily holds intermediate results while a program is executing.
- It reduces the waiting time of the CPU because the CPU can quickly read from and write to RAM.

RAM is temporary or volatile memory because its contents are lost when power is switched off. Even then it is essential because a computer needs a fast working memory during processing. Without RAM, every instruction and data item would have to be fetched from much slower storage devices, making the system extremely slow or unusable.

1(c) Explain the characteristics of I/O devices needed for the given situations.

Answer

Situation	Suitable device	Main characteristics
A printed page is to be converted to electronic form without retyping it.	Scanner with OCR software	Captures printed text/image optically. OCR converts scanned image of text into editable digital text. Resolution is measured in dpi.
A printed bill has a bar code to be read for payment details.	Bar-code reader/scanner	Reads black and white bar patterns using light sensor or laser; fast and accurate for billing, inventory and payment systems.
Very high-quality and high-precision graphs and pictures of different sizes are to be printed.	Plotter	Produces precise line drawings, engineering drawings, maps, banners and large-size graphics; suitable for CAD and technical diagrams.
A device is required to record voice.	Microphone	Converts sound waves into electrical/digital signals; used for voice recording, online meetings, speech input and multimedia applications.

1(d) What is the role of operating system? Explain memory management in operating system.

Answer

An operating system is system software that acts as an interface between user, application programs and computer hardware. It controls and coordinates all resources of the computer.

Main roles of an operating system:

- manages CPU, memory, files and input/output devices;
- provides user interface such as command line or graphical interface;
- loads and runs programs;

- controls security, user accounts and access permissions;
- handles errors and coordinates communication between hardware and software.

Memory management is the part of the operating system that controls main memory. It keeps track of which part of RAM is free and which part is occupied, allocates memory to programs, deallocates memory after use, protects one program's memory from another and supports virtual memory when RAM is insufficient. Efficient memory management allows several programs to run safely and smoothly at the same time.

1(e) Software needed to write and execute a C/C++ program.

Answer

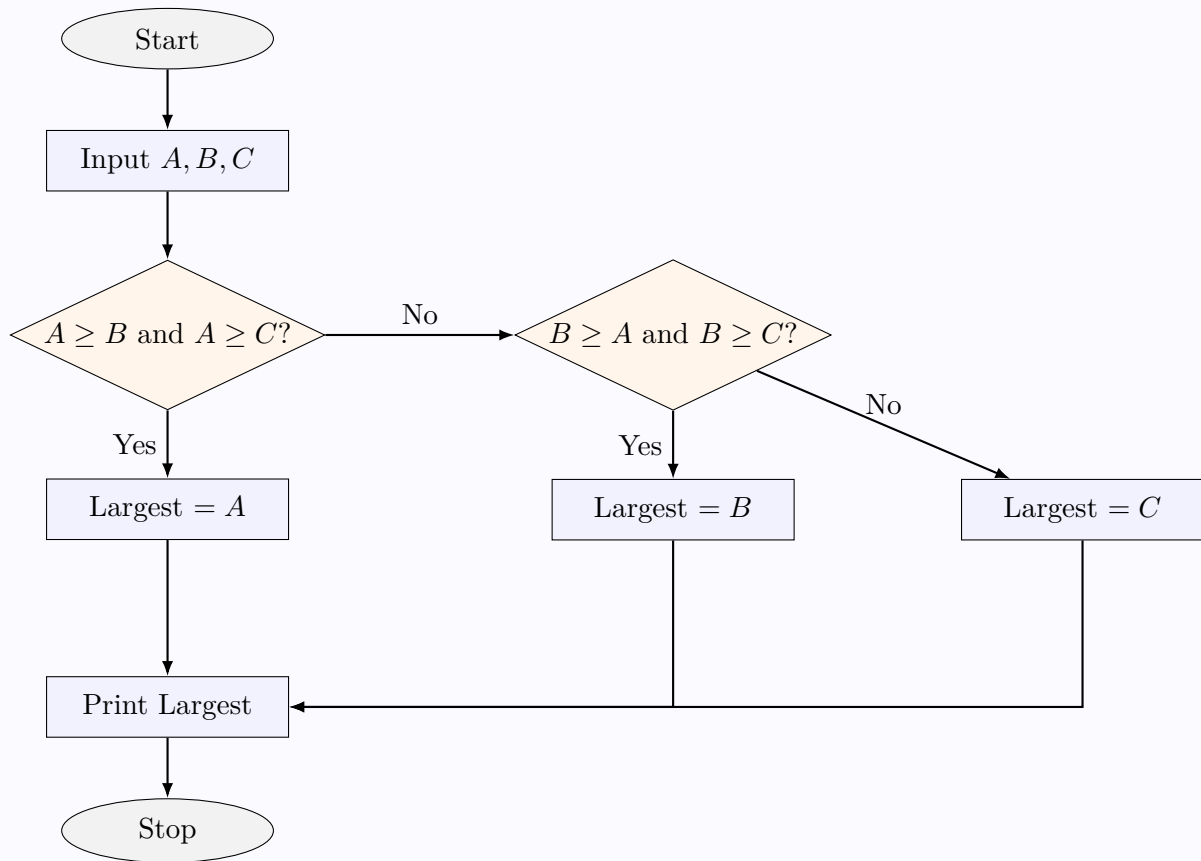
To write and execute a program in C or C++, the following software is needed:

Software	Features and role
Text editor or IDE	Used to type and edit source code. Examples: Notepad++, Code::Blocks, Visual Studio Code, Turbo C/C++ or Dev-C++. An IDE may include editor, compiler, debugger and project manager together.
Compiler	Translates the C/C++ source program into machine/object code. It checks syntax errors and produces object files. Examples: GCC, G++, Clang.
Linker	Combines object files and library files to create an executable program. It resolves references to functions such as <code>printf()</code> , <code>scanf()</code> or library routines.
Debugger	Helps in finding logical and runtime errors by running the program step by step, inspecting variables and locating faults.
Operating system	Provides file system, memory, CPU scheduling and execution environment required to run the program.

Typical sequence: write source code → compile → link → execute → debug if required.

1(f) Draw a flowchart to input three numbers and find the largest among them.

Answer

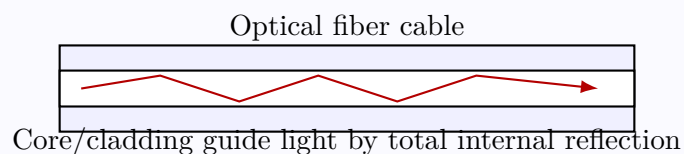


The flowchart first checks whether A is the largest. If not, it checks whether B is the largest. If both checks fail, C is the largest.

1(g) Characteristics, advantages and disadvantages of optical fiber cable.

Answer

Optical fiber cable is a transmission medium that carries data as light pulses through very thin strands of glass or plastic.



Characteristics:

- uses light instead of electrical signals;
- very high bandwidth and high speed;
- low signal loss over long distance;
- immune to electromagnetic interference;
- thin, light in weight and difficult to tap secretly.

Advantages: high data rate, long-distance communication, better security, low noise, low attenuation and suitability for internet backbones, LAN backbones and telephone networks.

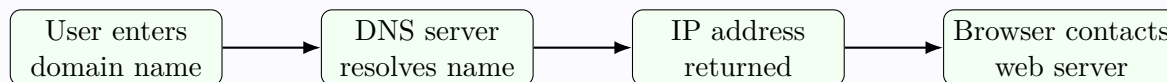
Disadvantages: costlier installation, fragile compared with copper cable, difficult splicing and repair, requires special connectors, optical transmitters and receivers.

1(h) What is the role of DNS? Explain with an example.

Answer

DNS stands for Domain Name System. Its role is to translate human-readable domain names into IP addresses understood by computers.

For example, when a user enters `www.ignou.ac.in` in a browser, the computer does not directly understand this name. It sends a DNS query to find the IP address of that domain. After the IP address is obtained, the browser can contact the correct web server.



Thus DNS acts like the phonebook of the internet. It allows users to remember names instead of numeric IP addresses.

1(i) Explain the role of any two folders that are part of your e-mail account.

Answer

Two common folders of an e-mail account are:

- **Inbox:** It stores received e-mails. New messages from other users usually appear in the Inbox unless filtered or marked as spam.
- **Sent Mail:** It stores copies of e-mails that the user has successfully sent. It helps in checking previous communication and proof of sending.

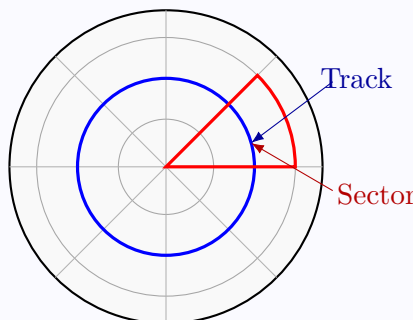
Other common folders include Drafts, Spam/Junk, Trash/Deleted Items and Archive. Drafts stores unfinished messages; Spam stores suspicious messages; Trash stores deleted messages temporarily.

2 Question 2

2(a) Explain data organization of a disk with the help of diagram(s).

Answer

A magnetic disk stores data on circular surfaces called platters. Each surface is divided into concentric circles called tracks. Each track is divided into sectors. A sector is the smallest physical unit that can be read or written. The set of tracks at the same position on all disk surfaces is called a cylinder.



Disk surface divided into tracks and sectors

Logical organization:

- Data is stored in files.

- Files are divided into blocks or clusters.
 - The file system maintains directories/folders and keeps track of which blocks belong to which file.
 - The operating system uses allocation tables or metadata to locate files on the disk.
- Therefore, disk organization has both physical structure (platter, track, sector, cylinder) and logical structure (file, folder, block/cluster and file-system metadata).

2(b) What is open source software? Explain open source development model.

Answer

Open source software is software whose source code is made available to users so that they can study, use, modify and distribute it under the terms of an open source licence.

Characteristics of open source software:

- source code is available;
- users can modify and improve the software;
- redistribution is allowed under licence conditions;
- development is often community based;
- bugs can be found and corrected by many contributors.

Open source development model:

1. A project releases its source code publicly.
2. Users, programmers and organizations study and test it.
3. Contributors report bugs, suggest features and submit code changes.
4. Maintainers review the changes for quality and security.
5. Accepted changes are merged into the main project.
6. New versions are released for public use.

Examples include Linux, LibreOffice, Firefox, Apache web server, Python and GCC.

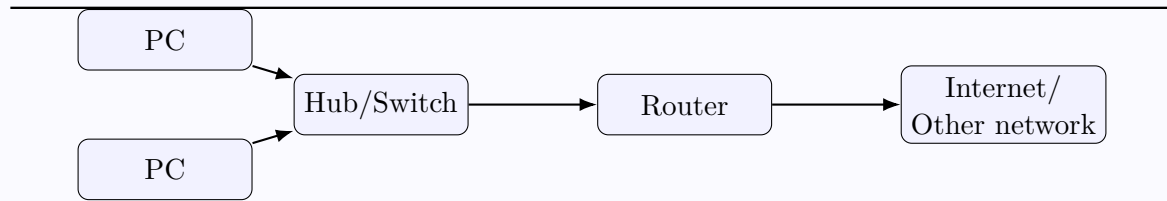
2(c) Explain the role of network interface card, modem, hub and router.

Answer

Device	Role
Network Interface Card (NIC)	Connects a computer to a network. It may be wired Ethernet or wireless Wi-Fi. It has a unique MAC address and handles sending/receiving network frames.
Modem	Modulates and demodulates signals. It converts digital computer data into a form suitable for transmission over telephone/cable/fiber line and converts received signals back into digital data.
Hub	A basic networking device that connects multiple computers in a LAN. It broadcasts incoming data to all connected ports, so it is less intelligent and less efficient than a switch.

Router

Connects two or more networks and forwards packets based on IP addresses. It is used to connect a LAN to another LAN or to the internet. It can perform routing, NAT, DHCP and firewall functions.



3 Question 3

3. Compare and contrast any five of the following.

Answer

All six comparisons are given for complete preparation.

(a) SRAM and DRAM

Point	SRAM	DRAM
Full form	Static RAM	Dynamic RAM
Storage cell	Uses flip-flop circuits	Uses capacitor and transistor
Refresh	Does not need regular refresh	Needs periodic refresh
Speed	Faster	Slower than SRAM
Cost	Costlier	Cheaper
Use	Cache memory	Main memory/RAM

(b) Parallel port and serial port

Point	Parallel port	Serial port
Data transfer	Sends multiple bits at a time	Sends one bit at a time
Cable	More wires	Fewer wires
Speed over distance	Good for short distance; signal issues over long distance	Better for longer distance
Traditional use	Printers, scanners	Modems, mouse, serial devices
Modern examples	Mostly replaced	USB and serial communication concepts remain common

(c) Subroutines and functions

Point	Subroutine	Function
Meaning	A named block of code used to perform a task	A named block of code that usually returns a value
Return value	May not return a value	Normally returns a value
Use	Reuse of repeated procedure	Computation and returning result
Example	Procedure to print a report	Function <code>sum(a,b)</code> returning <code>a+b</code>

(d) Word processing and spreadsheet software

Point	Word processor	Spreadsheet software
Main purpose	Preparing text documents	Organizing numeric/tabular data
Basic unit	Page/paragraph	Cell/row/column
Major features	Typing, formatting, spell check, mail merge	Formulas, functions, charts, sorting, filtering
Examples	MS Word, LibreOffice Writer	MS Excel, LibreOffice Calc

(e) Primary search engines and meta search engines

Point	Primary search engine	Meta search engine
Index	Maintains its own database/index	Does not maintain a full independent index
Working	Crawls and indexes web pages directly	Sends user query to several search engines and combines results
Control	More control over ranking	Depends on other search engines
Example	Google, Bing	Dogpile, MetaCrawler

(f) Microwave transmission and infrared transmission

Point	Microwave transmission	Infrared transmission
Medium	Electromagnetic waves in microwave range	Infrared light waves
Range	Suitable for long distance line-of-sight links	Short range, usually within a room
Obstacle effect	Needs line of sight; affected by buildings and weather	Cannot pass through walls
Use	Satellite, cellular backhaul, point-to-point links	Remote controls, short-range device communication

4 Question 4

4(a) What is collaboration in web applications? What is Wiki? Characteristics and uses of Wiki.

Answer

Collaboration in web applications means that two or more users can work together through the internet to create, edit, share or manage information. Examples include shared documents, discussion forums, online project management tools and wikis.

A **Wiki** is a web-based application that allows users to create, edit and link web pages collaboratively through a browser. Wikipedia is the most famous example.

Characteristics of Wiki:

- easy page creation and editing through browser;
- multiple users can contribute;
- pages are linked together using hyperlinks;
- version history is maintained;
- changes can be reviewed and reverted;
- supports collaborative knowledge building.

Activities for which Wiki can be used:

- preparing project documentation;
- creating institutional knowledge base;
- collaborative class notes;
- FAQs and help manuals;
- research group documentation;
- community information sharing.

4(b) What is cloud computing? Discuss key features.

Answer

Cloud computing is a model in which computing resources such as storage, servers, databases, software and processing power are provided through the internet as services. Users do not need to own or maintain all infrastructure locally.

Key features of cloud computing:

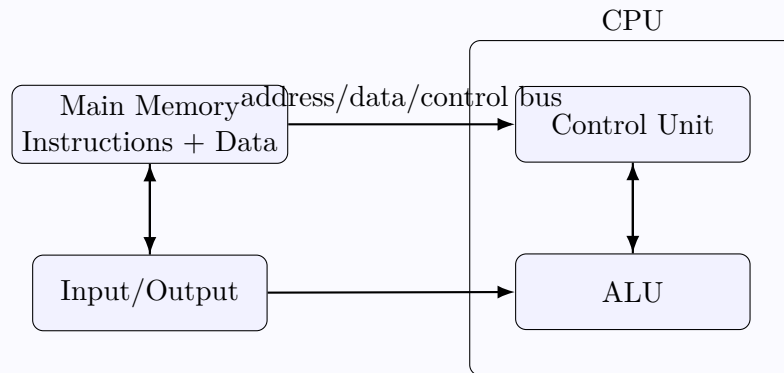
- **On-demand self-service:** users can use resources whenever needed.
- **Broad network access:** services are available through the internet from many devices.
- **Resource pooling:** resources are shared among many users.
- **Scalability:** resources can be increased or decreased according to demand.
- **Measured service:** usage can be monitored and billed according to consumption.
- **Reduced local maintenance:** hardware and software maintenance is handled by service provider.

Examples include Google Drive, Microsoft OneDrive, Gmail, AWS, Microsoft Azure and Google Cloud.

4(c) Explain Von Neumann architecture with the help of a diagram.

Answer

Von Neumann architecture is a computer architecture in which program instructions and data are stored in the same memory. The CPU fetches instructions from memory, decodes them and executes them one by one.



Main components:

- **Memory:** stores both data and instructions.
- **Control Unit:** controls fetching, decoding and execution of instructions.
- **ALU:** performs arithmetic and logical operations.
- **Input/Output units:** allow communication with external devices.
- **Buses:** transfer data, addresses and control signals.

The important feature is the stored-program concept: programs are stored in memory like data, so the same machine can perform different tasks by loading different programs.

4(d) What is an integrated circuit? Advantages of using integrated circuits.

Answer

An integrated circuit (IC) is a small semiconductor chip, usually made of silicon, on which many electronic components such as transistors, resistors, capacitors and diodes are fabricated together.

Advantages:

- very small size and light weight;
- high reliability because fewer external connections are needed;
- low power consumption;
- low cost in mass production;
- high processing speed;
- reduced heat generation and compact design.

ICs are the basis of modern processors, memory chips, controllers and almost all digital electronic devices.

5 Question 5

5(a) What is op-code in an instruction of a computer? Explain with example.

Answer

An instruction of a computer generally contains two parts: operation code and operand. The **op-code** tells the CPU what operation is to be performed. The operand tells on which data or memory location the operation is to be performed.

Example:

ADD R1, R2

Here, ADD is the op-code. It tells the CPU to perform addition. R1 and R2 are operands/registers involved in the operation.

Another example:

LOAD A, 5000

Here, LOAD is the op-code. It tells the CPU to load data from memory location 5000 into register A.

5(b) Give one example each of volatile and non-volatile memory.

Answer

- **Volatile memory:** RAM. Its contents are lost when power is switched off.
- **Non-volatile memory:** ROM, hard disk, SSD or flash memory. Its contents remain stored even when power is switched off.

5(c) What is the role of video card of a computer?

Answer

A video card, also called graphics card or display adapter, processes graphical data and sends the display signal to the monitor. It converts data from the CPU into images, text, video and graphics visible on screen.

Its main roles are:

- rendering 2D and 3D graphics;
- improving video and animation performance;
- supporting higher resolution and multiple monitors;
- reducing graphics load on the CPU, especially when it has a separate GPU.

5(d) Uses of mail merge feature of word processor with example.

Answer

Mail merge is a feature of a word processor that combines a standard document with a data source to create many personalized documents automatically.

Uses:

- sending personalized letters to many people;
- printing address labels and envelopes;
- generating certificates;
- preparing fee notices, appointment letters or invitation letters;
- creating e-mail messages with personalized names and details.

Example: A college wants to send the same notice to 500 students, but each notice must contain the student's name, roll number and course. The notice is typed once as the main document. A spreadsheet containing names, roll numbers and courses is used as the data source. Mail merge automatically produces 500 personalized notices.

5(e) Explain batch processing in the context of operating system.

Answer

Batch processing is a method in which similar jobs are collected together and executed as a group without continuous user interaction. The user submits jobs, and the operating system executes them one after another according to a schedule.

Features:

- suitable for large and repetitive tasks;
- no need for user interaction during execution;
- improves efficiency for routine jobs;
- output is produced after the batch is completed.

Examples include payroll processing, bank statement generation, electricity bill processing, examination result processing and bulk data conversion.

5(f) What is a browser? Features of browser. How does a browser communicate with web server?

Answer

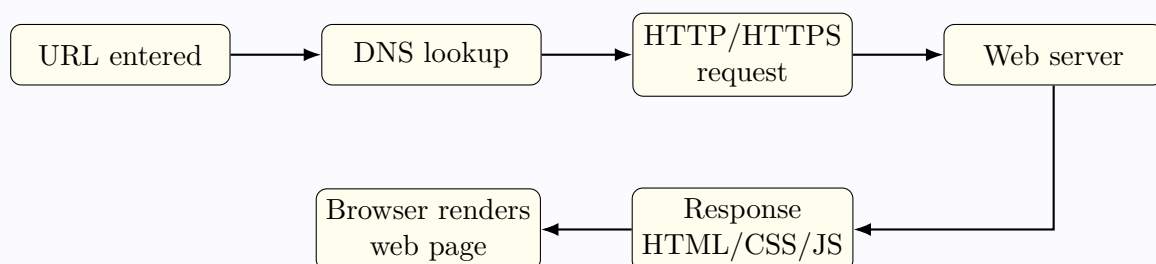
A browser is application software used to access, retrieve and display web pages and web applications on the internet. Examples are Google Chrome, Mozilla Firefox, Microsoft Edge and Safari.

Features of a browser:

- address bar to enter URL;
- support for HTML, CSS, JavaScript, images, audio and video;
- tabs for opening multiple pages;
- bookmarks/favourites;
- download manager;
- history and cache;
- security features such as HTTPS support, pop-up blocking and privacy controls.

Communication with web server:

1. User enters a URL such as `https://www.example.com`.
2. Browser uses DNS to find the IP address of the domain name.
3. Browser establishes connection with the web server, commonly using TCP/IP.
4. Browser sends an HTTP/HTTPS request to the server.
5. Server sends back an HTTP/HTTPS response containing HTML, CSS, images or other data.
6. Browser interprets the response and displays the web page.



Final Exam Note

For IGNOU answers, write point-wise, define the term first, then give features, examples and diagrams wherever asked. For number-system questions, show the calculation steps because marks are awarded for method as well as final answer.